While this pipeline is highly structured and achieves excellent recall (83.33%), the glaring bottleneck is the **low overall accuracy (54.72%) and precision (56.87%) on unseen subjects**. This indicates the model suffers from a high False Positive rate—it frequently misclassifies temporary facial movements (like a frown or a blink) as acute cognitive stress.

To elevate this architecture to a state-of-the-art (SOTA) level for a high-impact journal, you need to address the short temporal context and incorporate the physiological data you are currently leaving on the table. Here are the most impactful architectural modifications you can make:

## 1. Multi-Task Learning: Reintroduce the Missing rPPG Signal

You are using the UBFC-rPPG dataset but entirely bypassing its most valuable asset: the ground-truth physiological signals (Heart Rate and Blood Volume Pulse).

- **The Modification:** Instead of relying *only* on affective transfer (RAF-DB), add an auxiliary regression head to the Phase 3 BiGRU that predicts the continuous Heart Rate (HR) or Heart Rate Variability (HRV) alongside the binary stress classification.
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- **Why it works:** Cognitive stress is both affective (expressions) and physiological (heart rate). By forcing the model to predict HR, the shared layers learn to extract subtle skin-color variations (micro-blushes) alongside geometric muscle movements. This physiological grounding acts as a powerful regularizer that will drastically reduce your False Positives.
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## 2. Multi-Scale Temporal Windows (Addressing the 1.7s Limitation)

A 10-frame sliding window at 6 FPS captures only 1.7 seconds of data. Cognitive stress builds over time; a 1.7-second window is too short to distinguish between a momentary expression of confusion and sustained mental arithmetic stress.

- **The Modification:** Implement a **Two-Stream Temporal Network**.
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    - *Stream A (Micro):* Your current 1.7s window to catch sudden affective micro-expressions.
    - 
    - *Stream B (Macro):* A 10-second to 15-second window (60–90 frames) to capture macro-level changes in blinking rates, head pose stability, and physiological baselines.
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- **Why it works:** Concatenating the attention-pooled vectors from both streams gives the final classifier the context it needs to stop triggering "Stress" just because a subject frowned for one second.
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## 3. Upgrade Pre-Training: Spatial $\to$ Spatiotemporal Contrastive Learning

In Phase 1, you are using standard SimCLR, which applies spatial augmentations (cropping, color jitter) to static frames. It learns invariant *geometry* but ignores *motion*.

- **The Modification:** Upgrade from SimCLR to **VideoMoCo** or implement **Time-Contrastive Learning (TCN)**. Treat a sequence of frames (e.g., 5 frames) as the positive pair, forcing the model to learn that consecutive frames belong together.
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- **Why it works:** Stress is dynamic. If the backbone learns temporal consistency during self-supervised pre-training, it will provide much richer embeddings to the Phase 3 BiGRU, easing the burden on the recurrent layers.
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## 4. Upgrade the Deployment Filter (Beyond Moving Average)

Your Step 22 Moving-Average Smoothing Filter ($K=3$) reduced jitter by 34.6%, but a simple moving average introduces lag (latency) and dulls sharp, genuine stress onset signals.

- **The Modification:** Replace the K=3 moving average with a **1D Kalman Filter** or an **Exponential Moving Average (EMA)** parameterized by the model's output confidence score.
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- **Why it works:** A Kalman filter can predict the next state and update based on the model's new prediction. If the model outputs a sudden "Stress = 1" with low confidence, the filter smooths it out. If it outputs "Stress = 1" with 99% confidence, the filter updates instantly, eliminating lag while still reducing jitter.
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## The Highest ROI Recommendation

If you only have time to implement one change before submitting the manuscript, implement **Multi-Task Learning (Predicting Stress + HR)**. Because UBFC already has the ground-truth HR data, you do not need to label any new data. This single modification bridges affective computing with physiological sensing, which is exactly what reviewers at *IEEE Transactions on Affective Computing* look for.